Detection and Forensics Analysis

Module 2: Detection & Forensic Analysis

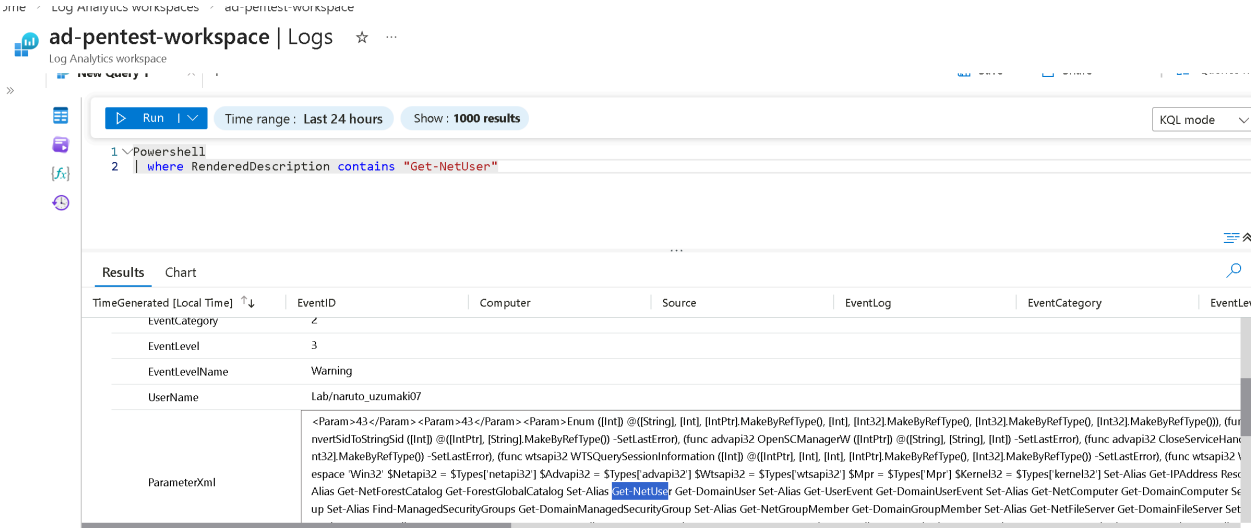
Methodology: SIEM Log Auditing & PowerShell Detection

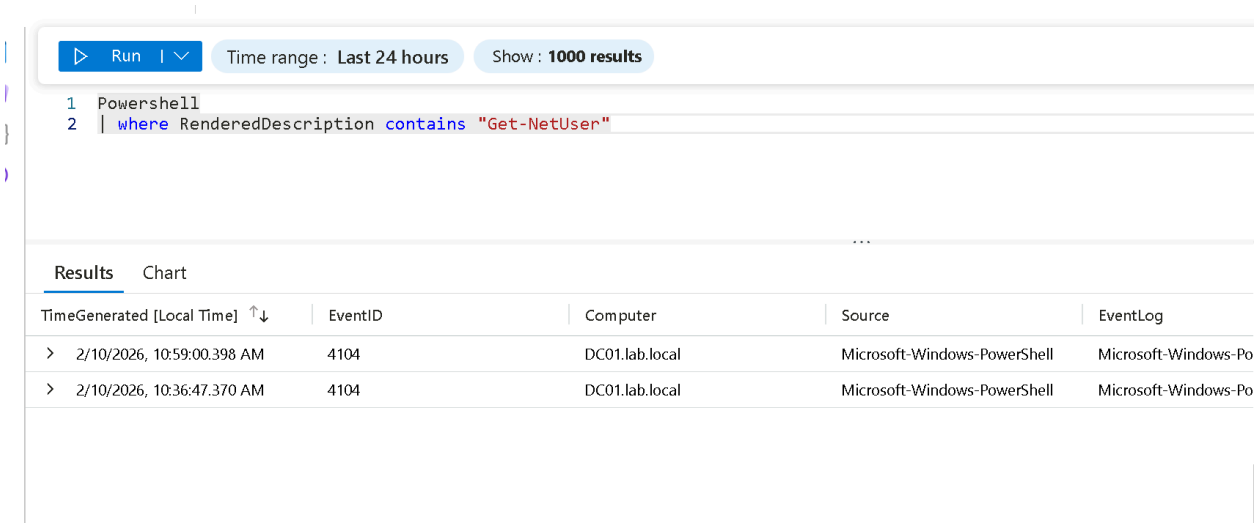
1. Overview

Attackers often rely on PowerShell-based tools like PowerView for enumeration. These leave a trail in the Windows Event Logs (Event ID 4104 - Script Block Logging), which can be ingested into a SIEM for centralized detection.

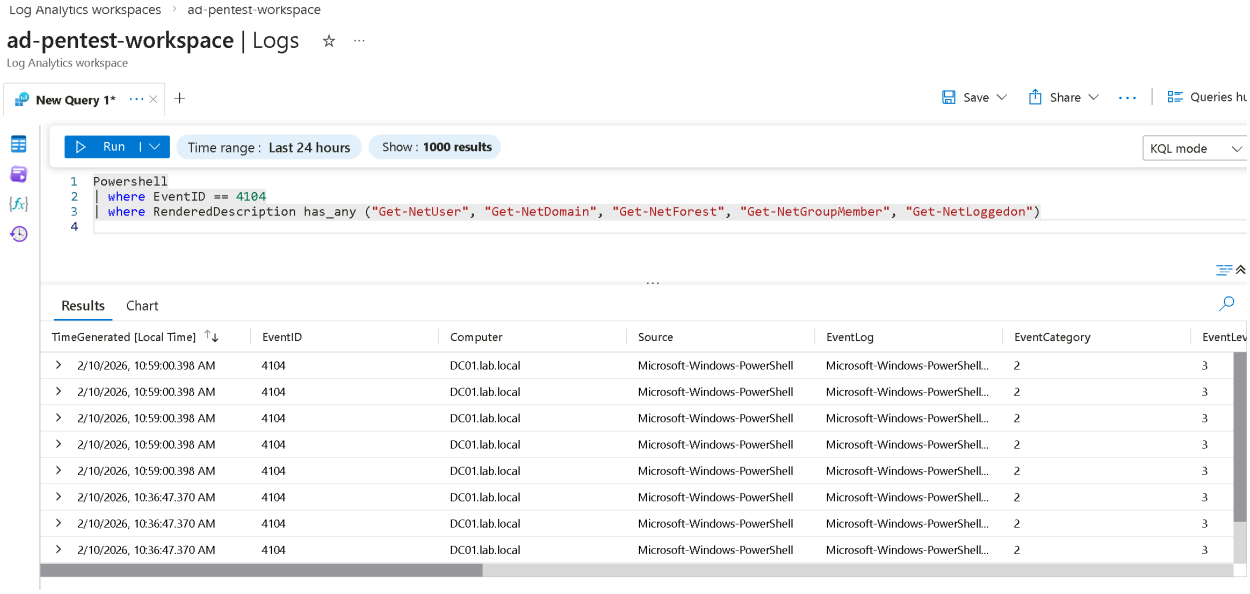
2. SIEM Query Analysis

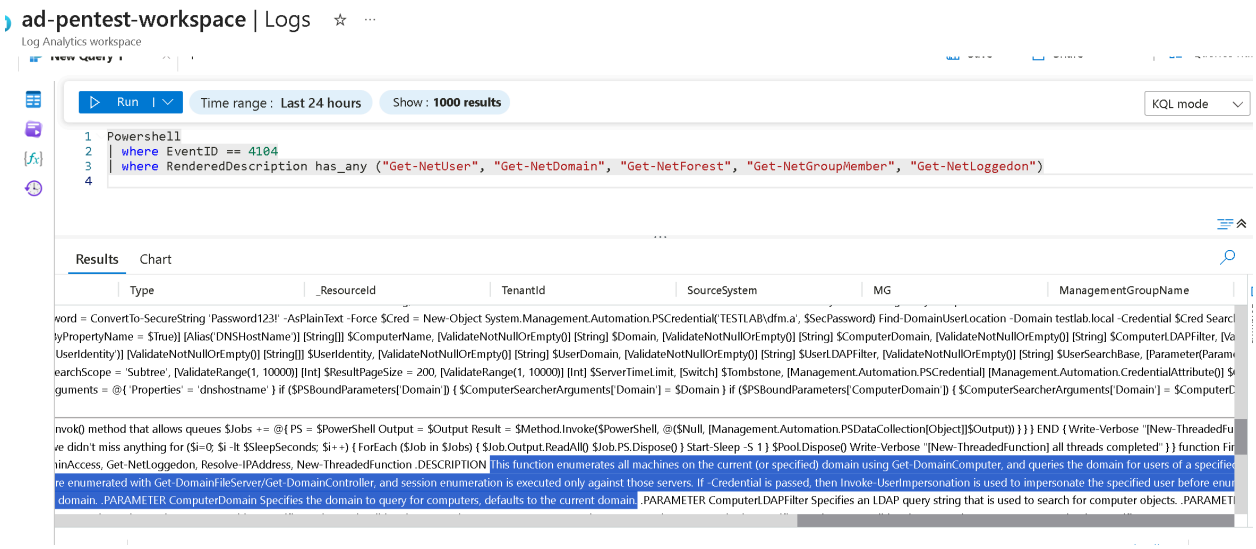
- **1.png & 2.png:** Initial log analysis identifies the user `Lab/naruto_uzumaki07` executing the `Get-NetUser` cmdlet, a core PowerView function used to harvest account details.



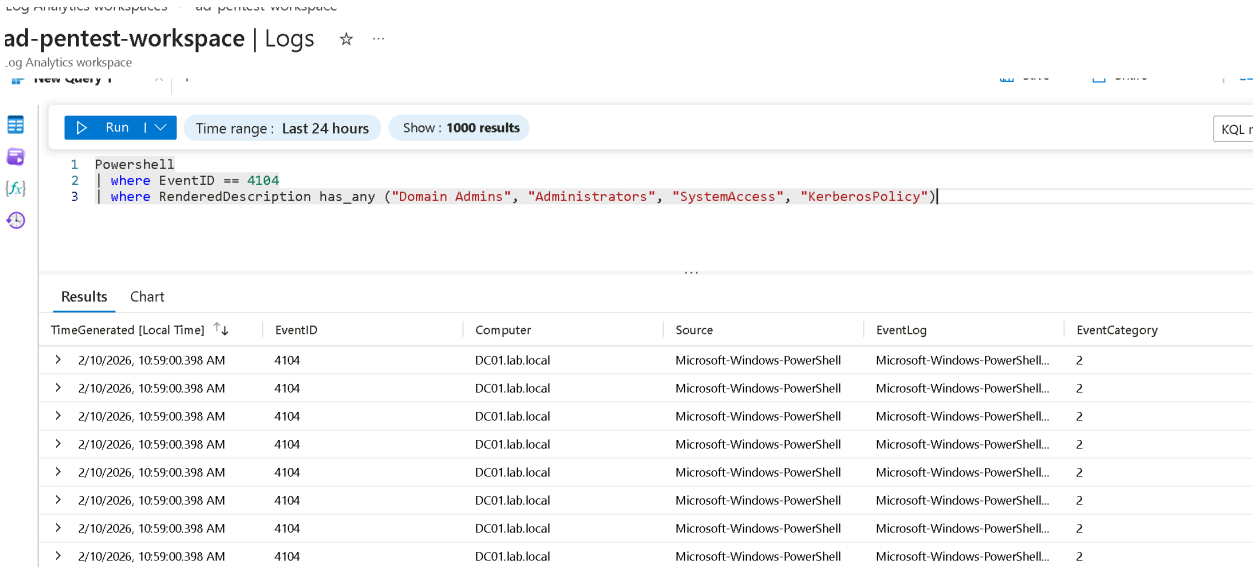


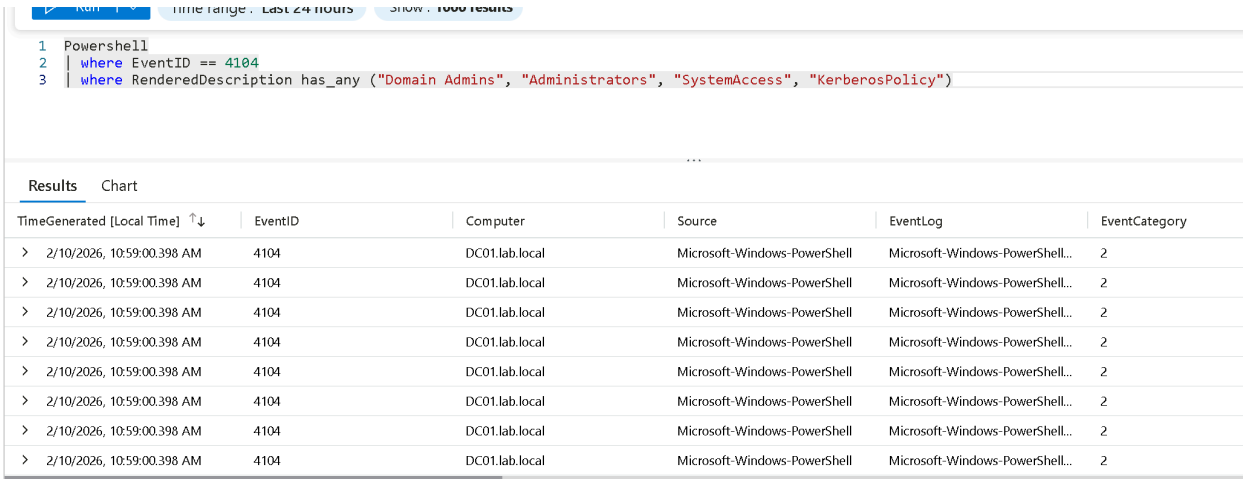
- **3.png & 4..png:** A broad query for multiple PowerView cmdlets (`Get-NetUser`, `Get-NetDomain`, `Get-NetForest`, etc.) confirms a sustained period of infrastructure discovery from the compromised host.



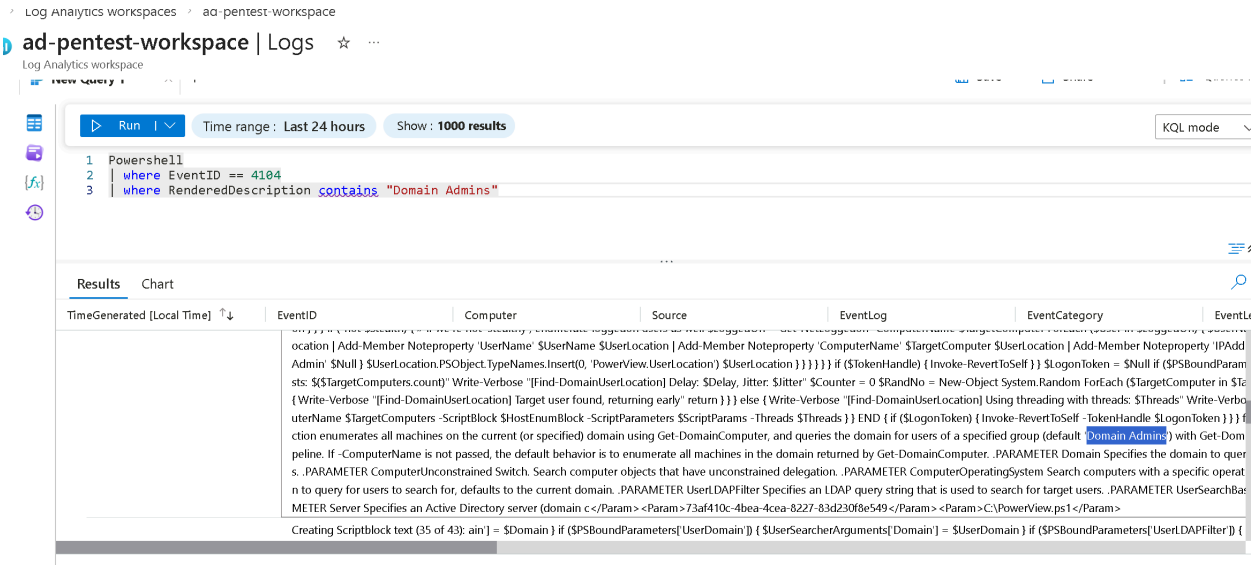


- **5.png & 7.png:** Using the `has_any` operator, we effectively filtered for high-value targets like "Domain Admins" and "Administrators," proving the attacker was actively mapping administrative groups.

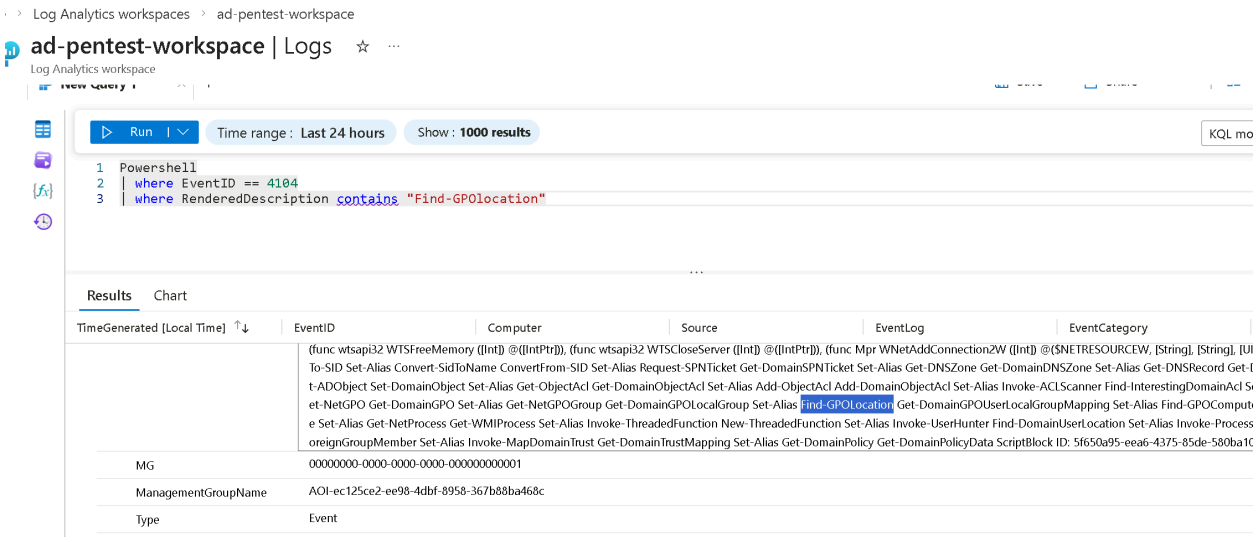


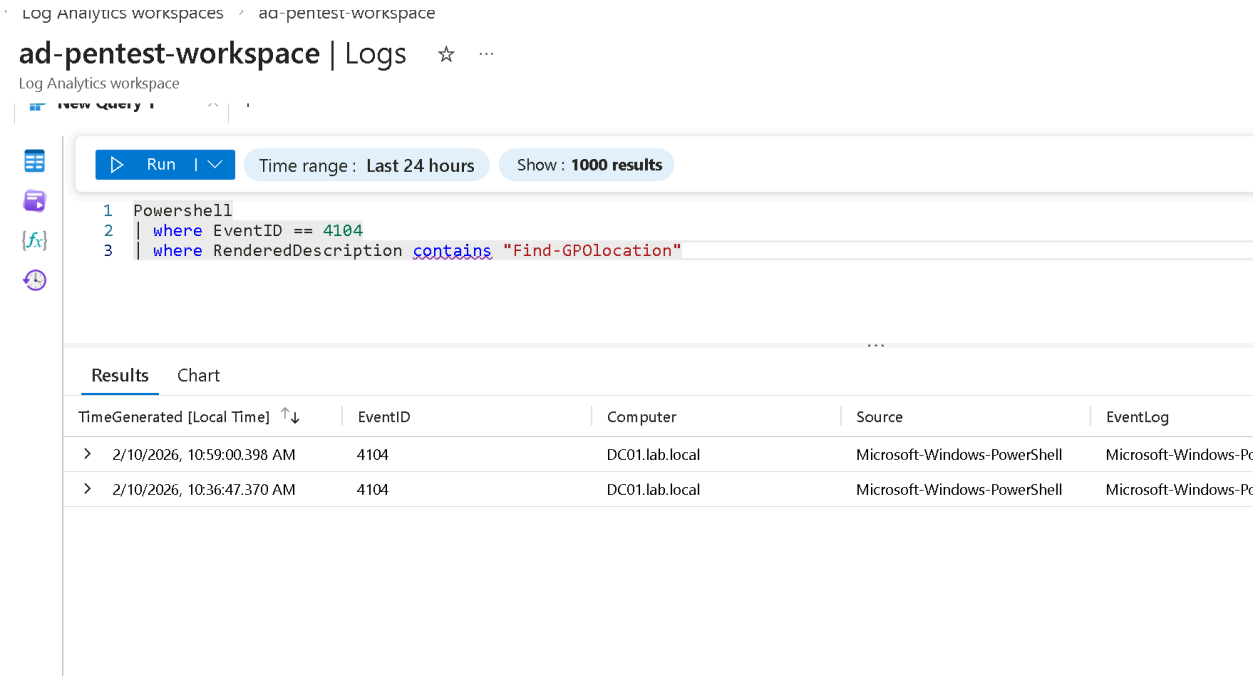


- **6.png:** Detailed inspection of the `RenderedDescription` field reveals the exact script block executed, including internal PowerView logic used to enumerate members of the "Domain Admins" group.

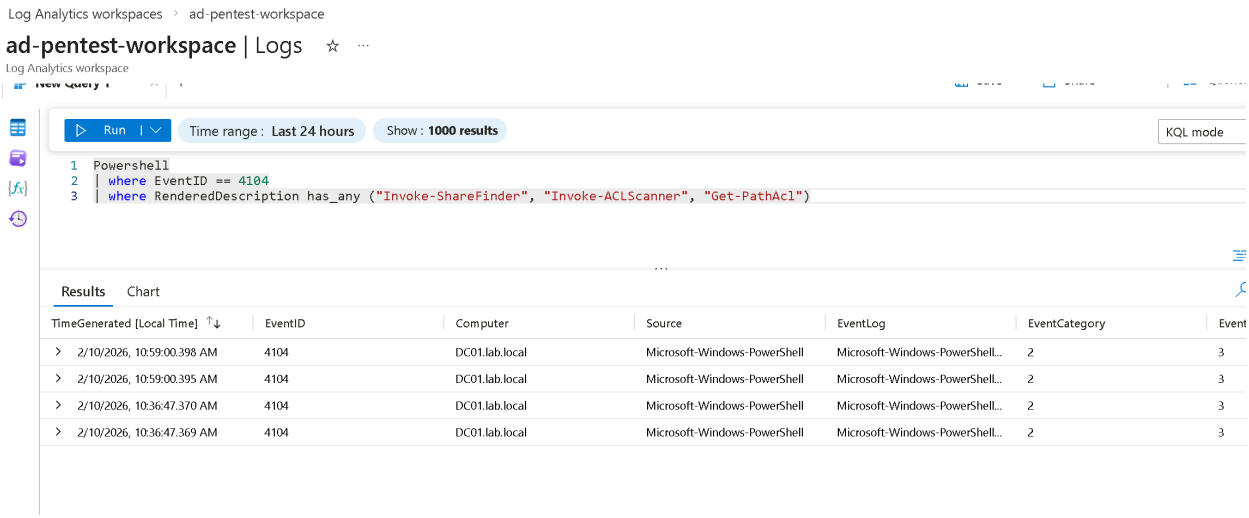


- **8.png & 9.png:** Identification of the `Find-GPOLocation` command indicates the attacker was investigating the GPO structure to find potential misconfigurations or sensitive files.





- **10.png:** Centralized analysis of Event ID 4104 logs on `DC01.lab.local` shows the timeline of reconnaissance, specifically flagging dangerous tools like `Invoke-ShareFinder` and `Invoke-ACLScanner`.

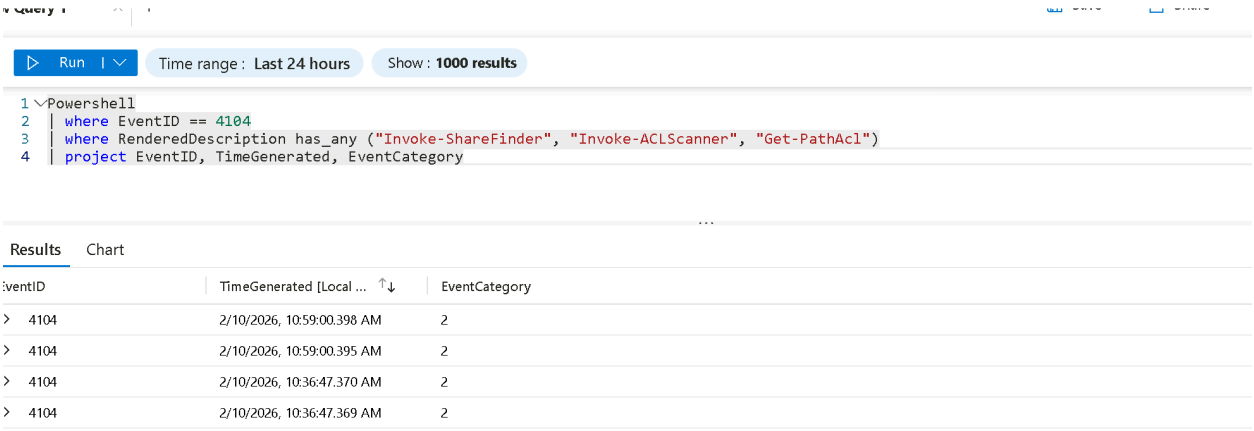
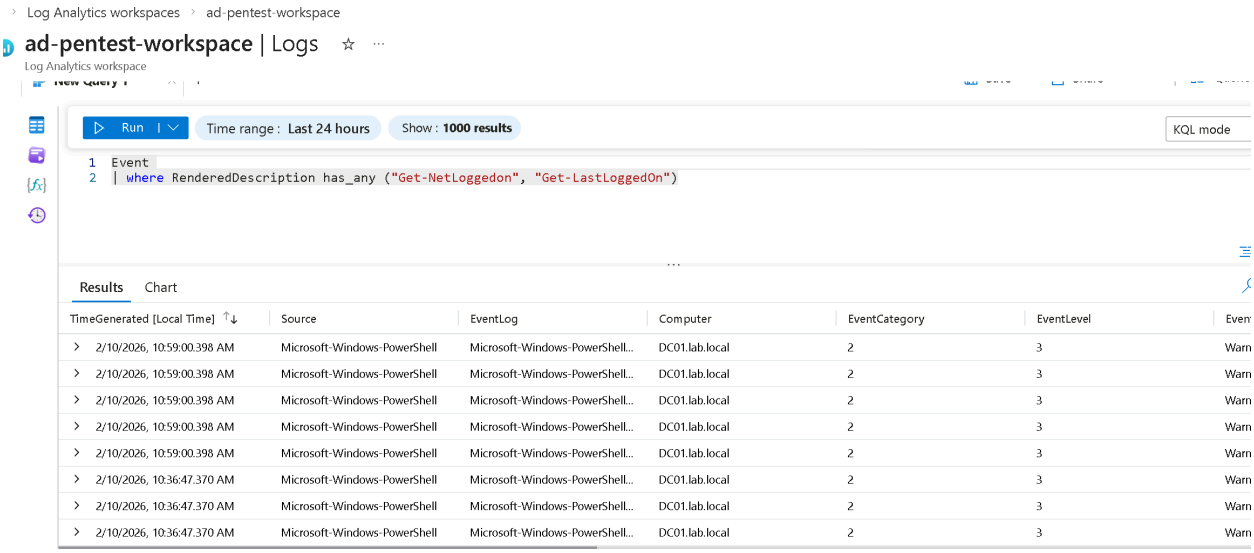


Detection Point	Forensic Indicator (KQL Query)	Significance
Cmdlet Execution	`RenderedDescription contains "Get-NetUser"`	Direct evidence of PowerView usage.
Privileged Targeting	`has_any ("Domain Admins", "Administrators")`	Shows intent to escalate to Tier Zero.
Share Discovery	`RenderedDescription has_any ("Invoke-ShareFinder")`	Indicates a search for sensitive data (e.g., "Confidential" share).
User Account	`UserName == "Lab/naruto_uzumaki07"`	Attributes the activity to the compromised credential.

Security Recommendations

- **Enable Script Block Logging:** Ensure PowerShell Operational logging is enabled domain-wide via GPO to capture the full command syntax.
- **Alert on PowerView Cmdlets:** Implement real-time SIEM alerts for high-fidelity PowerView keywords (e.g., `Invoke-ACLScanner`, `Invoke-ShareFinder`).
- **Monitor for 4104 Events:** Centralize all Script Block events to a SIEM for long-term storage and correlation against known administrative hostnames.
- **Enforce PowerShell Constrained Language Mode:** Where possible, limit the PowerShell capabilities for non-administrative users to prevent script execution.

11.png: This query filters for **EventID 4104** and uses the `has_any` operator to specifically alert on high-fidelity PowerView functions like `Invoke-ShareFinder`, `Invoke-ACLScanner`, and `Get-PathAcl`. It provides a clear timeline of when the attacker began scanning for network shares and abusable ACLs on the domain.



12.png: A broader hunting query targeting session enumeration cmdlets such as `Get-NetLoggedon` and `Get-LastLoggedOn`. Detecting these commands is critical as they indicate an attacker is actively profiling user activity to identify high-value targets for credential harvesting or token impersonation.